What Drives Income Shifting by Multinational Enterprises?

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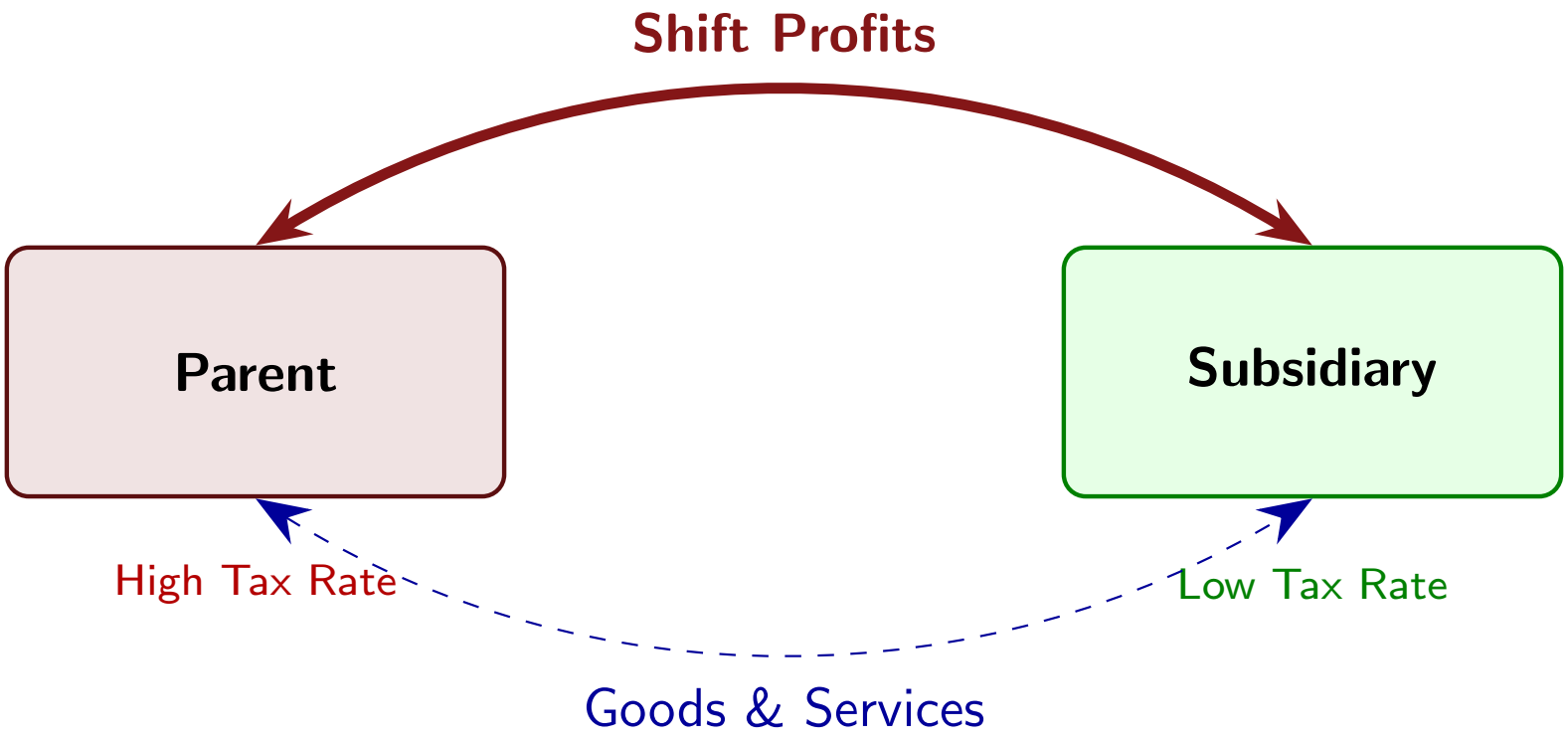
Income Shifting – Example 1

HQ sells product to subsidiary at a low transfer price ($40), shifting $60 of profit to the low-tax jurisdiction.

Income Shifting – Example 2

HQ raises the transfer price to $80, keeping $60 of profit in the low-tax jurisdiction.

What is Income Shifting?



By adjusting **transfer prices** on intercompany transactions, MNEs can shift taxable income to lower-tax jurisdictions.

Why Should We Care?

Legalized Tax Fraud: How Top US Corporations Continue to Profit Through Offshore Tax Havens

February 18, 2015

Are Big Companies Really Moving $100 Billion to Tax Havens?

DUKE FUQUA MARCH 19, 2025

At a glance

$492 billion is lost to tax havens a year

Of the $492 billion lost a year...

$347.6 billion is lost to cross-border corporate tax abuse by multinational corporations

144.8 billion is lost to offshore tax evasion by wealthy individuals

Global tax abuse harms everybody

Higher income countries lose bigger sums, but lower income countries' losses make up a bigger share

5x

Lower income countries lose 5 times as much, as a share of their public health budgets, compared to higher income

The State of Tax Justice 2024

TAX JUSTICE NETWORK

Research Motivation

Understanding the drivers of income shifting is the key to constructing productive tax incentives which will influence the critical tradeoff between:

Shareholder Value

Minimizing taxes paid to maximize after-tax profits and shareholder returns.

Tax Responsibility

Contributing appropriate tax amounts to fund government services and infrastructure.

This study extends prior work by:

- Covering fiscal years **2010–2024** (excluding 2017)
  - Including pre- and post-TCJA periods

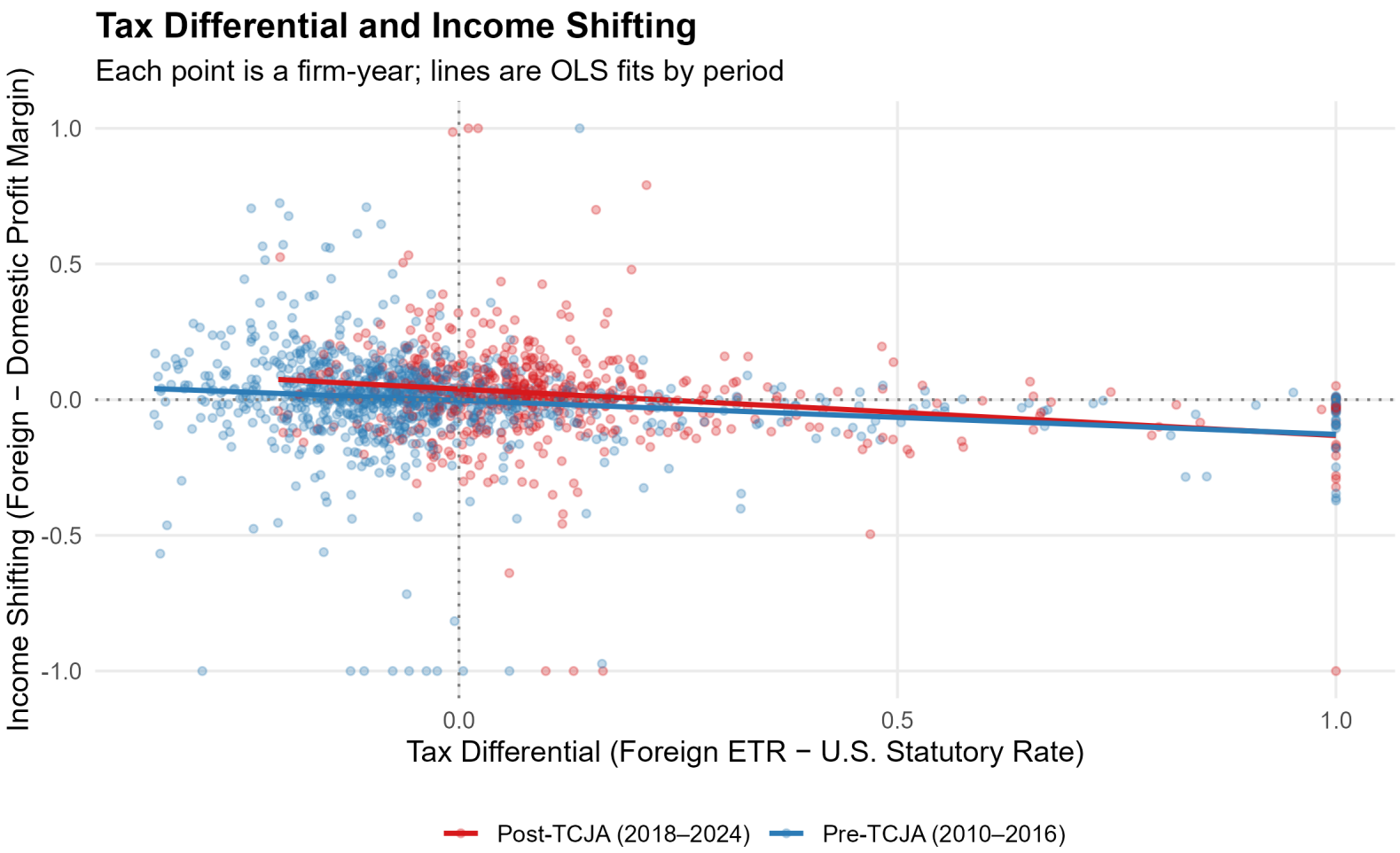
Data

- **Source:** WRDS Compustat (Fundamentals Annual + Geographic Segments)
  - Publicly traded U.S. multinational companies
- **Sample:** 1,553 firm-year observations from 346 U.S. MNEs (2010–2024, excluding 2017)

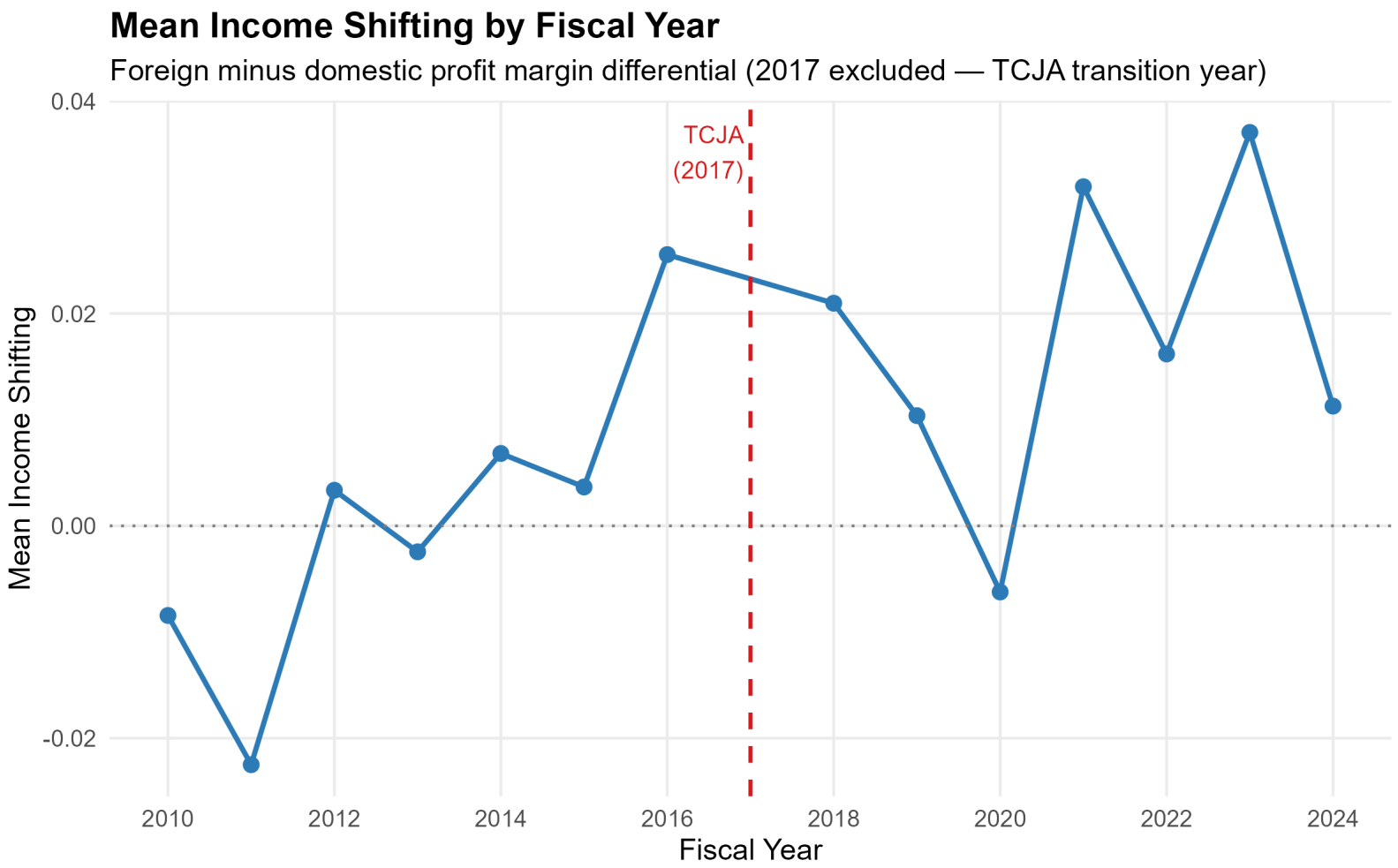
Variable Definitions

Variable	Definition
income_shift	Foreign profit margin − domestic profit margin
tax_diff	Foreign effective tax rate − U.S. statutory rate
ww_profit	Worldwide pre-tax income / total revenue
size	Natural log of total assets
leverage	Total debt / total assets
rnd_intensity	R&D expense / total revenue
ad_intensity	Advertising expense / total revenue
ppe_intensity	Net PP&E / total assets
year_trend	Fiscal year − 2010
post_tcja	$= 1$ if fiscal year ≥ 2018, $= 0$ otherwise

Data – continued



Data – continued



Data – continued

Descriptive Statistics

Variable	N	Mean	Median	Std Dev
income_shift	1,553	0.0065	0.0097	0.1802
tax_diff	1,553	0.0355	−0.0222	0.2512
ww_profit	1,553	0.1175	0.0970	0.0929
size	1,553	8.2410	8.1890	1.8920
leverage	1,553	0.2870	0.2520	0.2210
rnd_intensity	1,553	0.0310	0.0060	0.0590
ad_intensity	1,553	0.0120	0.0000	0.0330
ppe_intensity	1,553	0.2100	0.1380	0.2040

Empirical Methods

Regression Equation:

$$\text{income_shift}_{it} = \beta_0 + \beta_1 \text{tax_diff}_{it} + \beta_2 \text{ww_profit}_{it} + \boldsymbol{\beta}\mathbf{X}_{it} + \varepsilon_{it}$$

Progressive specifications:

- **Model 1:** Baseline with all controls (robust SEs)
- **Model 2:** Year fixed effects
- **Model 3:** TCJA interaction terms
- **Model 4:** Industry (2-digit SIC) fixed effects

Controls ($\mathbf{X}_{it}$): firm size, leverage, R&D intensity, advertising intensity, PPE intensity, year trend, post-TCJA indicator

Sensitivity: Re-estimated on a restricted sample (995 obs) excluding ambiguous geographic segments

Regression Results (Model 1)

Variable	Estimate	Std Error	t Stat	p-value
Intercept	0.108	0.024	4.50	<.0001
tax_diff	−0.162	0.027	−6.00	<.0001
ww_profit	−0.469	0.069	−6.80	<.0001
size	0.005	0.003	1.97	0.0488
leverage	0.086	0.022	3.91	<.0001
rnd_intensity	−0.092	0.102	−0.90	0.3655
ad_intensity	0.231	0.092	2.51	0.0118
ppe_intensity	0.047	0.035	1.34	0.1791
year_trend	0.002	0.002	1.03	0.3018
post_tcja	0.017	0.018	0.94	0.3473
$F = 23.41$, $p < .0001$		Adj. $R^2 = 11.61\%$, $N = 1{,}536$		

Findings – continued

Key Findings

Tax Differential ($\beta = -0.162$, $p < .0001$):
Higher foreign tax rates are associated with less outward income shifting. This is consistent with profit-maximizing behavior under the TCJA's reduced U.S. rate.

Worldwide Profitability ($\beta = -0.469$, $p < .0001$):
More profitable firms shift less income abroad, suggesting these firms face greater scrutiny or have less incentive to shift.

Leverage ($\beta = 0.086$, $p < .0001$):
More leveraged firms shift more income, suggesting firms use debt financing and transfer pricing simultaneously as tax strategies.

Firm Size ($\beta = 0.005$, $p = .049$):
Larger firms engage in marginally more shifting, consistent with having the resources and infrastructure to do so.

Other Controls: R&D intensity, PPE intensity, year trend, and post-TCJA indicator are not statistically significant.

What Did We Learn?

- Tax differentials are the **strongest driver** of income shifting — a 1 pp increase in the foreign-domestic tax gap reduces outward shifting by 0.16 pp
- Results are robust across year fixed effects, TCJA interactions, industry fixed effects, and a restricted sensitivity sample
- Post-TCJA coefficient is insignificant — **no clear evidence the reform reduced shifting**

Policy Implication

Statutory rate cuts alone may not curb profit shifting. Other tax rules or incentives should be considered including different enforcement.

Limitations: Compustat geographic segments may not capture all shifting channels; excludes financial firms

Thank You!

Any Questions?